(5.20) $\beta - \sqrt{\beta^2 - \omega_0^2}$ $\downarrow$ as $\beta \uparrow$ for $\beta > \omega_0$ & sketch

$$\frac{d}{d\beta}\left[\beta - \sqrt{\beta^2-\omega_0^2}\right] = 1 - \frac{1}{2}(\beta^2-\omega_0^2)^{1/2}\, 2\beta = 1 - \frac{\beta}{\sqrt{\beta^2-\omega_0^2}}$$

we know $\beta > \omega_0$

$\beta^2 > \omega_0^2$

$\sqrt{\beta^2-\omega_0^2} > 0$ ✓

$\omega_0^2 > 0$

$0 > -\omega_0^2$

$\beta^2 > \beta^2 - \omega_0^2$

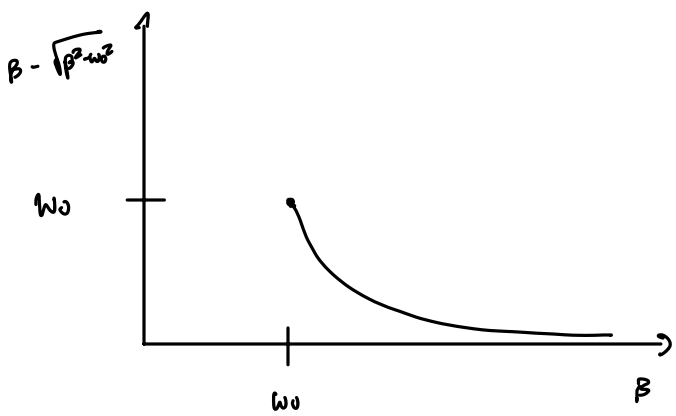
$\sqrt{\beta^2} > \sqrt{\beta^2-\omega_0^2}$ (since both sides are positive)

$\beta > \sqrt{\beta^2-\omega_0^2}$

$\frac{\beta}{\sqrt{\beta^2-\omega_0^2}} > 1$

$0 > 1 - \frac{\beta}{\sqrt{\beta^2-\omega_0^2}} =$

$0 > \frac{d}{d\beta}\left[\beta - \sqrt{\beta^2-\omega_0^2}\right] \Rightarrow \beta - \sqrt{\beta^2-\omega_0^2}$ is decreasing with β



(5.24) $x_1(t) = e^{-\beta t}\cos(\omega_1 t)$ $\xrightarrow[\beta \to \omega_0]{\text{Show:}}$ $x_1(t) = e^{-\beta t}$

$x_2(t) = e^{-\beta t}\sin(\omega_1 t)$ $\qquad$ $x_2(t) \to te^{-\beta t}$

$x_1(t) = e^{-\beta t}\cos(\omega_1 t)$, $\quad \omega_1 = \sqrt{\omega_0^2-\beta^2}$

$\to e^{-\beta t}\cos\left(\sqrt{\omega_0^2-\omega_0^2}\, t\right) = e^{-\beta t}\cos(0) = e^{-\beta t}$ ☑

$x_2(t) = e^{-\beta t}\sin\omega_1 t \to 0$

try $\frac{x_2(t)}{\omega_1} \to \frac{e^{-\beta t}\sin\left(\sqrt{\omega_0^2-\omega_0^2}\, t\right)}{\sqrt{\omega_0^2-\omega_0^2}} \to \frac{0}{0}$ use L'Hôpital's rule

$$\lim_{\beta\to 0} \frac{e^{-\beta t}\sin\left(\sqrt{\omega_0^2-\beta^2}\,t\right)}{\sqrt{\omega_0^2-\beta^2}} = \lim_{\beta\to\omega_0} \frac{\frac{\partial}{\partial\beta}\left[e^{-\beta t}\sin\left(\sqrt{\omega_0^2-\beta^2}\,t\right)\right]}{\frac{\partial}{\partial\beta}\left[\sqrt{\omega_0^2-\beta^2}\right]}$$

$$= \lim_{\beta\to\omega_0} \frac{-te^{-\beta t}\sin\left(\sqrt{\omega_0^2-\beta^2}\,t\right) + e^{-\beta t}\cos\left(\sqrt{\omega_0^2-\beta^2}\,t\right)\frac{1}{2}t\,(\omega_0^2-\beta^2)^{-1/2}(-2\beta)}{\frac{1}{2}(\omega_0^2-\beta^2)^{-1/2}(-2\beta)}$$

$$= \lim_{\beta\to\omega_0} \frac{-te^{-\beta t}\sin\left(\sqrt{\omega_0^2-\beta^2}\,t\right) - t\beta(\omega_0^2-\beta^2)^{-1/2}e^{-\beta t}\cos\left(\sqrt{\omega_0^2-\beta^2}\,t\right)}{-\beta(\omega_0^2-\beta^2)^{-1/2}}$$

$$= \lim_{\beta\to\omega_0} +\frac{t}{\beta}\sqrt{\omega_0^2-\beta^2}\,e^{-\beta t}\sin\left(\sqrt{\omega_0^2-\beta^2}\,t\right) + te^{-\beta t}\cos\left(\sqrt{\omega_0^2-\beta^2}\,t\right)$$

$$\to \frac{t}{\beta}(0)e^{-\beta t}(0) + te^{-\beta t}\cos(0) = 0 + te^{-\beta t}$$

$$= te^{-\beta t} \quad \checkmark$$

5.30 $\quad x(t) = C_1 e^{-(\beta-\sqrt{\beta^2-\omega_0^2})t} + C_2 e^{-(\beta+\sqrt{\beta^2-\omega_0^2})t}$

a) $C_1(x_0, v_0)$, $C_2(x_0, v_0) = ?$

$$x_0 = C_1 e^0 + C_2 e^0 = C_1 + C_2$$

$$v_0 = \dot{x}(t=0)$$

$$\dot{x}(t) = -C_1(\beta-\sqrt{\beta^2-\omega_0^2})e^{-(\beta-\sqrt{\beta^2-\omega_0^2}\,t)} - C_2(\beta+\sqrt{\beta^2-\omega_0^2})e^{-(\beta+\sqrt{\beta^2-\omega_0^2})t}$$

$$v_0 = -C_1(\beta-\sqrt{\beta^2-\omega_0^2})e^0 - C_2(\beta+\sqrt{\beta^2-\omega_0^2})e^0$$

$$= -C_1\beta - C_1\sqrt{\beta^2-\omega_0^2} - C_2\beta + C_2\sqrt{\beta^2-\omega_0^2}$$

$$v_0 = -\underbrace{(C_1+C_2)}_{=x_0}\beta + (C_2-C_1)\sqrt{\beta^2-\omega_0^2}$$

$$v_0 = -x_0\beta + (C_1-C_2)\sqrt{\beta^2-\omega_0^2}$$

$$\left.\begin{aligned} \frac{v_0+x_0\beta}{\sqrt{\beta^2-\omega_0^2}} &= C_1-C_2 \\ x_0 &= C_1+C_2 \end{aligned}\right\}$$

add together $\to \dfrac{v_0+x_0\beta}{\sqrt{\beta^2-\omega_0^2}} + x_0 = 2C_1$

$$\boxed{\frac{1}{2}\left(x_0 + \frac{v_0+x_0\beta}{\sqrt{\beta^2-\omega_0^2}}\right) = C_1}$$

subtract

$\Downarrow$

$$x_0 - \left(\frac{v_0+x_0\beta}{\sqrt{\beta^2-\omega_0^2}}\right) = 2C_2$$

$$\boxed{\frac{1}{2}\left(x_0 - \frac{v_0+x_0\beta}{\sqrt{\beta^2-\omega_0^2}}\right) = C_2}$$

b) Sketch $x(t)\big|_{x_0=0}$, $x(t)\big|_{v_0=0}$

$$x(t)\Big|_{x_0=0} = C_1\Big|_{x_0=0} e^{-(\beta-\sqrt{\beta^2-\omega_0^2})t} + C_2\Big|_{x_0=0} e^{-(\beta+\sqrt{\beta^2-\omega_0^2})t}$$

$$C_1\Big|_{x_0=0} = \frac{v_0}{2\sqrt{\beta^2-\omega_0^2}} \qquad C_2\Big|_{x_0=0} = \frac{-v_0}{2\sqrt{\beta^2-\omega_0^2}}$$

$$x(t)\Big|_{x_0=0} = \frac{v_0}{2\sqrt{\beta^2-\omega_0^2}}\left(e^{-(\beta-\sqrt{\beta^2-\omega_0^2})t} - e^{-(\beta+\sqrt{\beta^2-\omega_0^2})t}\right)$$

$$\omega_0^2>0 \Rightarrow \beta^2-\omega_0^2<\beta^2$$

$$\sqrt{\beta^2-\omega_0^2}<\beta$$

$$0<\beta-\sqrt{\beta^2-\omega_0^2} \Rightarrow e^{-(\beta-\sqrt{\beta^2-\omega_0^2})t} \text{ is exponential decay}$$

$$0<\underbrace{\beta}_{\text{positive}}+\underbrace{\sqrt{\beta^2-\omega_0^2}}_{\text{positive}} \Rightarrow e^{-(\beta+\sqrt{\beta^2-\omega_0^2})t} \text{ is exponential decay}$$

Which dominates at late times?

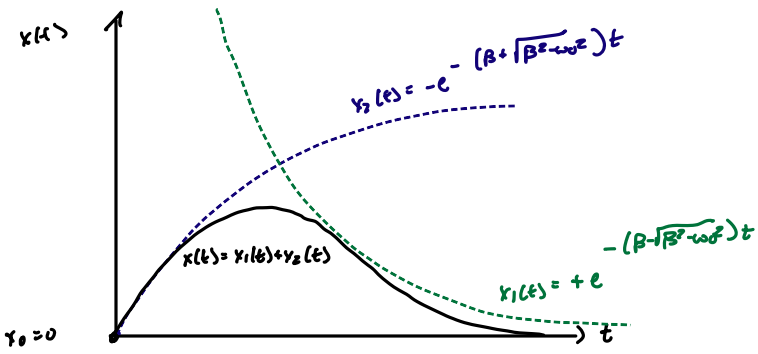
$$\sqrt{\beta^2-\omega_0^2}>0$$

$$\beta+\sqrt{\beta^2-\omega_0^2}>\beta>\beta-\sqrt{\beta^2-\omega_0^2}$$

$$\Rightarrow \beta+\sqrt{\beta^2-\omega_0^2} > \beta-\sqrt{\beta^2-\omega_0^2} \Rightarrow \begin{array}{ll} e^{-(\beta-\sqrt{\beta^2-\omega_0^2})t} & \text{decays more slowly} \\ e^{-(\beta+\sqrt{\beta^2-\omega_0^2})t} & \text{decays faster} \end{array}$$

$$x(t)\Big|_{x_0=0} = \frac{v_0}{2\sqrt{\beta^2-\omega_0^2}}\left(\underbrace{e^{-(\beta-\sqrt{\beta^2-\omega_0^2})t}}_{x_1(t)} - \underbrace{e^{-(\beta+\sqrt{\beta^2-\omega_0^2})t}}_{x_2(t)}\right)$$

$x_1(t)$ more important at late times



$\Rightarrow$ Sketch of $x(t)\big|_{x_0=0}$

Next, $v_0 = 0$:

$$x(t)\Big|_{v_0=0} = C_1\Big|_{v_0=0} e^{-(\beta-\sqrt{\beta^2-\omega_0^2})t} + C_2\Big|_{v_0=0} e^{-(\beta+\sqrt{\beta^2-\omega_0^2})t}$$

$$C_1\Big|_{v_0=0} = \frac{1}{2}\left(x_0 + \frac{x_0\beta}{\sqrt{\beta^2-\omega_0^2}}\right) = \frac{x_0}{2}\left(1+\frac{\beta}{\sqrt{\beta^2-\omega_0^2}}\right)$$

$$C_2\Big|_{v_0=0} = \frac{1}{2}\left(x_0 - \frac{x_0\beta}{\sqrt{\beta^2-\omega_0^2}}\right) = \frac{x_0}{2}\left(1-\frac{\beta}{\sqrt{\beta^2-\omega_0^2}}\right)$$

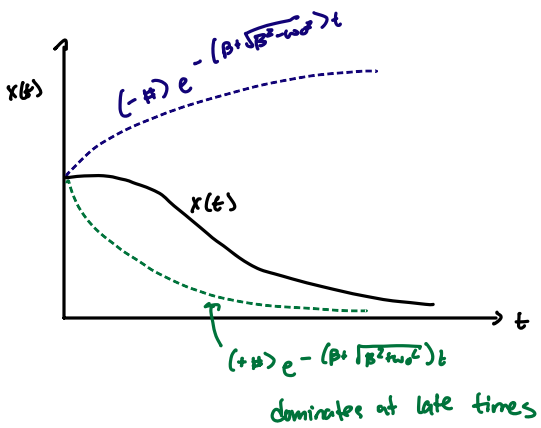
$$x(t)\Big|_{v_0=0} = \frac{x_0}{2}\left[\left(1+\frac{\beta}{\sqrt{\beta^2-\omega_0^2}}\right)e^{-(\beta-\sqrt{\beta^2-\omega_0^2})t} + \left(1-\frac{\beta}{\sqrt{\beta^2-\omega_0^2}}\right)e^{-(\beta+\sqrt{\beta^2-\omega_0^2})t}\right]$$

$$\beta^2 > \beta^2-\omega_0^2$$

$$\beta > \sqrt{\beta^2-\omega_0^2}$$

$$\frac{\beta}{\sqrt{\beta^2-\omega_0^2}} > 1 \quad \Rightarrow C_1 > 0, \quad C_2 < 0 \quad \text{as before}$$

the difference is slope = 0 initially



Sketch for $v_0 = 0$

c) $\beta \to 0$, $x(t) \to x(t)$ underdamped

$$x(t) = C_1 e^{-(\beta-\sqrt{\beta^2-\omega_0^2})t} + C_2 e^{-(\beta+\sqrt{\beta^2-\omega_0^2})t}$$

$$\lim_{\beta\to 0} C_1 = \frac{1}{2}x_0 + \frac{v_0}{2\sqrt{-\omega_0^2}} = \frac{1}{2}x_0 + \frac{v_0}{2i\omega_0} = \frac{1}{2}x_0 - \frac{iv_0}{2\omega_0}$$

$$\lim_{\beta\to 0} C_2 = \frac{1}{2}x_0 - \frac{v_0}{2\sqrt{-\omega_0^2}} = \frac{1}{2}x_0 + \frac{iv_0}{2\omega_0}$$

$$\lim_{\beta\to 0} x(t) = \left(\frac{1}{2}x_0 - \frac{iv_0}{2\omega_0}\right)e^{-(-\sqrt{-\omega_0^2})t} + \left(\frac{1}{2}x_0 + \frac{iv_0}{2\omega_0}\right)e^{-\sqrt{-\omega_0^2}\,t}$$

$$\lim_{\beta\to 0} x(t) = \left(\frac{1}{2}x_0 - \frac{iv_0}{2\omega_0}\right)e^{i\omega_0 t} + \left(\frac{1}{2}x_0 + \frac{iv_0}{2\omega_0}\right)e^{-i\omega_0 t}$$

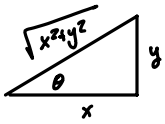
$$= \frac{1}{2}x_0\left(e^{i\omega_0 t} + e^{-i\omega_0 t}\right) - \frac{iv_0}{\omega_0}\left(e^{i\omega_0 t} - e^{-i\omega_0 t}\right)$$

$$= \frac{1}{2}x_0\,(2\cos\omega_0 t) - \frac{iv_0}{2\omega_0}\,(2i\sin\omega_0 t)$$

$$= x_0\cos\omega_0 t + \frac{v_0}{\omega_0}\sin\omega_0 t \quad \Rightarrow \text{ undamped soln} \quad ☑$$

5.35 a) $z = x+iy \Rightarrow z = re^{i\theta}$

$$z = x+iy = \sqrt{x^2+y^2}\left[\frac{x}{\sqrt{x^2+y^2}} + i\,\frac{y}{\sqrt{x^2+y^2}}\right] = \sqrt{x^2+y^2}\left[\cos\theta + i\sin\theta\right]$$



where $\theta \in \mathbb{R}$

$$= \sqrt{x^2+y^2}\;e^{i\theta} \equiv re^{i\theta} \quad ☑$$

(b) $|z| = r \Rightarrow |z|^2 = zz^*$

$$z = re^{i\theta} = r(\cos\theta + i\sin\theta)$$

$$\Rightarrow z^* = r(\cos\theta - i\sin\theta) = r\left[\cos\theta + i\sin(-\theta)\right] = re^{-i\theta}$$

$$zz^* = re^{i\theta}\,re^{-i\theta} = r^2 e^{i\theta - i\theta} = r^2 e^0 = r^2 = |z|^2 \quad ☑$$

(c) $z^* = e^{-i\theta}$ proven above

(d) $(zw)^* = z^* w^*$

$\left(\frac{1}{z}\right)^* = \frac{1}{z^*}$

$$(zw)^* = \left(r_1 e^{i\theta_1} r_2 e^{i\theta_2}\right)^* = \left[r_1 r_2 e^{i(\theta_1+\theta_2)}\right]^* = r_1 r_2 e^{-i(\theta_1+\theta_2)} = r_1 e^{-i\theta_1} r_2 e^{-i\theta_2} = z^* w^* \quad ☑$$

$$\left(\frac{1}{z}\right)^* = \left(\frac{1}{re^{i\theta}}\right)^* = \left(\frac{1}{r}e^{-i\theta}\right)^* = \frac{1}{r}e^{+i\theta} = \frac{1}{re^{-i\theta}} = \frac{1}{z^*} \quad ☑$$

(e) $z = \frac{a}{b+ic} \Rightarrow |z|^2 = \frac{a^2}{b^2+c^2}$

$$z = \frac{a}{b+ic} \qquad z^* = a^*\left(\frac{1}{b+ic}\right)^* = a^*\frac{1}{(b+ic)^*} = \frac{a}{b-ic}$$

$$zz^* = \frac{a}{b+ic}\,\frac{a}{b-ic} = \frac{a^2}{b^2 + icb - bic - i^2c^2} = \frac{a^2}{b^2+c^2} \quad ☑$$

5.44 $\omega = \omega_0$

a) Show $E = \frac{1}{2} m \omega^2 A^2$

$E = T + U$

when $x = A$, $\dot{x} = 0 \Rightarrow T = 0$

$$E = U\Big|_{x=A} = \frac{1}{2} K A^2 = \frac{1}{2} m \omega_0^2 A^2 = \frac{1}{2} m \omega^2 A^2 \quad \checkmark$$

$\omega_0^2 = \frac{K}{m}$

b) Show $\Delta E_{dis} = 2\pi m \beta \omega A^2$

$$\Delta E_{dis} = \int dE_{dis} = \int \frac{dE_{dis}}{dt} dt = -\int_0^\tau F v \, dt \quad , \quad F = -bv$$

$$= \int_0^\tau b v^2 dt \qquad x = A\cos(\omega t - \delta) \qquad v = -A\omega(\sin \omega t - \delta)$$

$$= \int_0^\tau b A^2 \omega^2 \sin^2(\omega t - \delta) dt = b A^2 \omega^2 \int_0^\tau \sin^2(\omega t - \delta) dt$$

$$= b A^2 \omega^2 \left[\frac{\omega t - \delta}{2\omega} + \frac{1}{4\omega} \sin(2\omega t - 2\delta) \right]_0^\tau$$

$$= b A^2 \omega^2 \left[\frac{\omega \tau - \delta}{2\omega} + \frac{1}{4\omega} \sin(2\omega\tau - 2\delta) + \frac{\delta}{2\omega} - \frac{1}{4\omega} \sin(-2\delta) \right] \qquad \tau = \frac{2\pi}{\omega}$$

$\sin(4\pi + x) = \sin x$

$$= b A^2 \omega^2 \left[\frac{2\pi - \delta}{2\omega} + \cancel{\frac{1}{4\omega} \sin(4\pi - 2\delta)} + \frac{\delta}{2\omega} - \cancel{\frac{1}{4\omega} \sin(-2\delta)} \right]$$

$$= b A^2 \omega^2 \left[\frac{\pi}{\omega} - \frac{\delta}{2\omega} + \frac{\delta}{2\omega} \right] = \pi b A^2 \omega = \pi \, 2m\beta A^2 \omega = 2\pi m \beta \omega A^2 \quad \checkmark$$

c) Show $Q = 2\pi \frac{E}{\Delta E_{dis}}$

$$\frac{E}{\Delta E_{dis}} = \frac{\frac{1}{2} \cancel{m} \omega^2 \cancel{A^2}}{2\pi \cancel{m} \beta \cancel{A^2} \omega} = \frac{\omega}{4\pi\beta} = \frac{Q}{2\pi} \quad \checkmark$$

(5.47) Prove $\displaystyle\int_{-\tau/2}^{\tau/2} \cos n\omega t \cos m\omega t \, dt = \begin{cases} \tau/2 & m=n\neq 0 \\ 0 & m\neq n \end{cases}$

$$\int_{-\tau/2}^{\tau/2} \cos n\omega t \sin m\omega t = 0$$

$\displaystyle\int_{-\tau/2}^{\tau/2} \cos n\omega t \sin m\omega t = 0$ because $\cos n\omega t$ = even, $\sin n\omega t$ = odd

full integrand is odd with equal & opposite bounds of integration

$\displaystyle\int_{-\tau/2}^{\tau/2} \cos n\omega t \cos m\omega t \, dt$ use trig ID: $\cos a \cos b = \dfrac{\cos(a+b) + \cos(a-b)}{2}$

$$= \frac{1}{2}\int_{-\tau/2}^{\tau/2} \cos\left[(n+m)\omega t\right] dt + \frac{1}{2}\int_{-\tau/2}^{\tau/2} \cos\left[(n-m)\omega t\right] dt = I_1 + I_2$$

$$I_1 = \frac{1}{2}\,\frac{1}{(n+m)\omega} \sin\left[(n+m)\omega t\right]\Big|_{-\tau/2}^{\tau/2} = \frac{1}{2(n+m)\omega}\left\{\sin\left[(n+m)\,\omega\,\frac{\tau}{2}\right] - \sin\left[-(n+m)\,\omega\,\frac{\tau}{2}\right]\right\}$$

Note, for the Fourier series n, m are always ≥ 0 so $n+m \neq 0$ so long as we avoid $n=m=0$

$$= \frac{2\sin\left[(n+m)\,\omega\tau/2\right]}{2(n+m)\omega} = \frac{\sin\left[(n+m)\,2\pi/2\right]}{(n+m)\omega} = \frac{\sin\left(\pi(n+m)\right)}{(n+m)\omega} = 0$$

$(\omega\tau = 2\pi)$

$$I_2 = \begin{cases} \frac{1}{2}\,\frac{1}{(n-m)\omega} \sin\left[(n-m)\omega t\right]_{-\tau/2}^{\tau/2} & n\neq m \\ \frac{1}{2}\displaystyle\int_{-\tau/2}^{\tau/2} \cos(0)\,dt = \frac{1}{2}\int_{-\tau/2}^{\tau/2} dt & n=m \end{cases}$$

$$I_2 = \begin{cases} \frac{1}{2}\,\frac{1}{(n-m)\omega}\left\{2\sin\left[(n-m)\,\omega\tau/2\right]\right\} = \dfrac{\sin\left[(n-m)\,2\pi/2\right]}{(n-m)\omega} = \dfrac{\sin\left[\pi(n-m)\right]}{(n-m)\omega} = 0 & n\neq m \\ \frac{1}{2}\, t\Big|_{-\tau/2}^{\tau/2} = \frac{1}{2}\left(\tau/2 + \tau/2\right) = \frac{1}{2}\tau & n=m \end{cases}$$

$$\Rightarrow \quad I_1 + I_2 = \begin{cases} 0 & n\neq m \\ \frac{1}{2}\tau & n=m\neq 0 \end{cases}$$

☑

(5.51) Show $z_n = \sum_{n=0}^{\infty} c_n e^{in\omega t}$, $c_n = \dfrac{f_n}{\omega_0^2 - n^2\omega^2 + 2i\beta\omega n}$, $g(t) = \sum_{n=0}^{\infty} f_n e^{in\omega t}$

$x_n = Re(z_n)$

Solve: $Dz = \ddot{z} + 2\beta\dot{z} + \omega_0^2 z = g$

$$Dz = g$$
$$Re[Dz] = Re[g]$$

$$D[Re\, z] = f$$
$$Dx = f$$

Choose $z(t) = \sum_{n=0}^{\infty} c_n e^{in\omega t}$.

Then $Dz = \sum_{n=0}^{\infty} c_n (in\omega)^2 e^{in\omega t} + 2\beta c_n (in\omega) e^{in\omega t} + \omega_0^2 c_n e^{in\omega t} = \sum_{n=0}^{\infty} f_n e^{in\omega t}$

$\Rightarrow \quad c_n\left[-n^2\omega^2 + 2i\beta n\omega + \omega_0^2\right] = f_n \qquad \forall n$

$\Rightarrow \quad c_n = \dfrac{f_n}{\omega_0^2 - n^2\omega^2 + 2i\beta n\omega}$ ☑